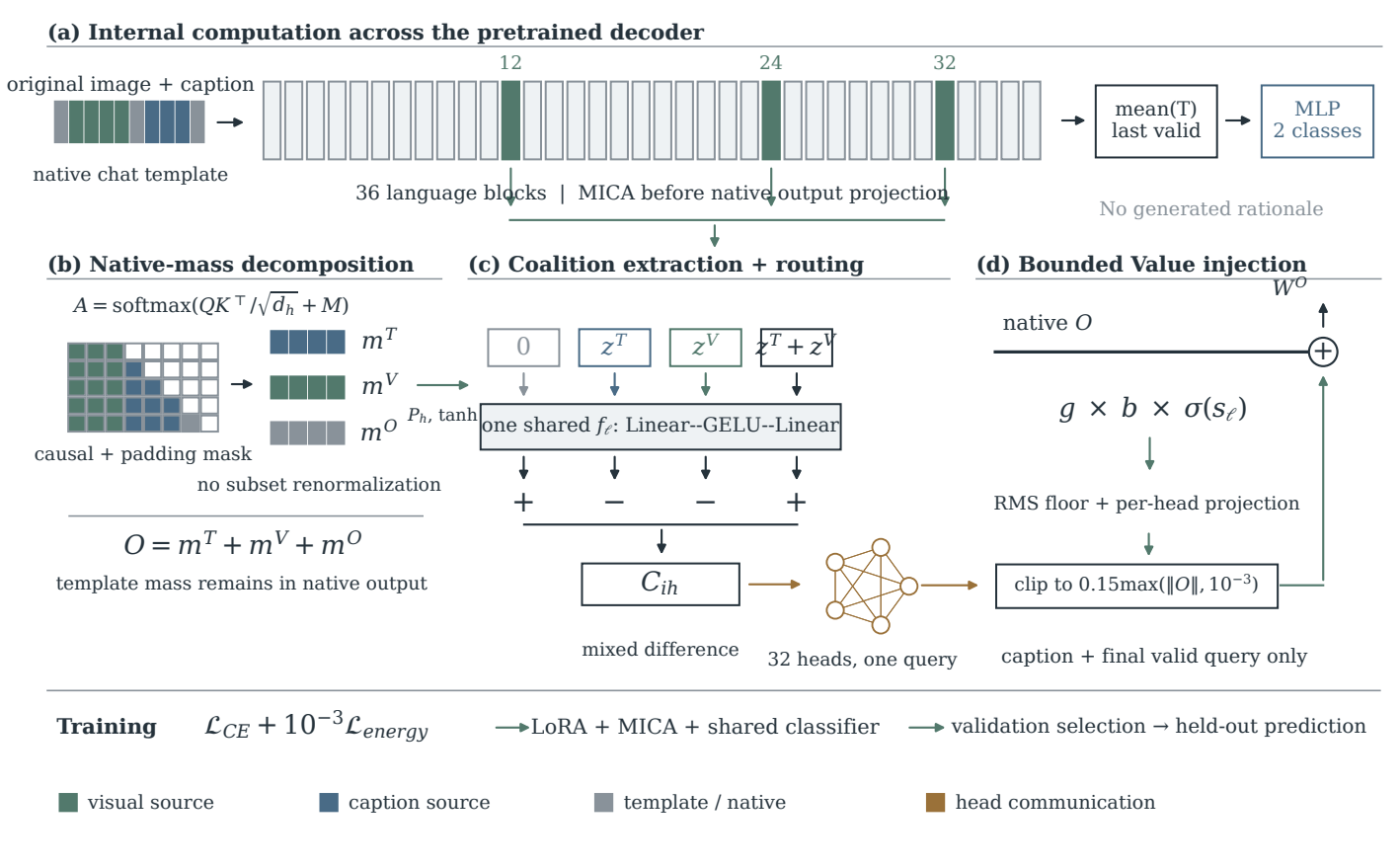
(a) Internal computation across the pretrained decoder
original image + caption
native chat template
12
24
32
36 language blocks | MICA before native output projection
mean(T)
last valid
MLP
2 classes
No generated rationale
(b) Native-mass decomposition
$A = \mathrm{softmax}(QK^\top/\sqrt{d_h} + M)$
causal + padding mask
m^T
m^V
m^O
no subset renormalization
$O = m^T + m^V + m^O$
template mass remains in native output
(c) Coalition extraction + routing
P_h, tanh
0
z^T
z^V
$z^T + z^V$
one shared f_ℓ: Linear--GELU--Linear
+ − − +
C_{ih}
mixed difference
32 heads, one query
(d) Bounded Value injection
W^O
native O
$g \times b \times \sigma(s_\ell)$
RMS floor + per-head projection
clip to $0.15\max(\|O\|, 10^{-3})$
caption + final valid query only
Training
$\mathcal{L}_{CE} + 10^{-3}\mathcal{L}_{energy}$
LoRA + MICA + shared classifier
validation selection → held-out prediction
visual source
caption source
template / native
head communication